

MOHAMED A.K.I. AHMED

Aerospace Engineering



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SPECIALIZATION OVERVIEW

Gas Turbine Engine Design; Propulsion Systems Analysis; Aerothermal & CFD Modeling; Multidisciplinary Design Optimization (MDO); Numerical Methods & Simulation; Optimization, Sensitivity & Uncertainty Analysis

EDUCATION

Masters of Applied Science in Aerospace Engineering

2025

École de technologie supérieure (ÉTS), Montreal, Quebec, Canada

- Transitioned from Fast-track-to-PhD to Master's program.
- Conducted research at Pratt & Whitney Canada.
- Thesis Title: "Surrogate-based optimization for evaluating turbine cooling impacts on aero-engine performance"
- Link to Thesis: [Press Here](#)

Bachelor of Applied Science in Aerospace Engineering (Honours)

2021

Middle East Technical University (METU), Ankara, Turkey

WORK EXPERIENCE

Pratt & Whitney Canada (Longueuil, Quebec, Canada)

Aug 2021 - Aug 2025

Mitacs Fellowship Research Student

- Member of the Numerical Methods group (Research & Technology), focusing on gas turbine multi-disciplinary optimization during preliminary design.
- Developed a Preliminary MDO framework that uses Design of Experiments (DOE) and Surrogate Assisted Optimization (SAO) techniques to optimize turbine design for minimum SFC with integrated cooling models.
- Collaborated with Advanced Performance, Turbine Aerodynamics, and Durability teams to develop an engine-level MDO framework.
- Developed modular, reusable Python/MATLAB tools with version control and automated validation/testing to ensure reliability and deployment across engineering teams.
- Supported fellow research students through mentorship, training, and collaboration on numerical methods projects.

Teaching

Turbomachinery & Propulsion

Jan 2024 - Mar 2024

- Developed and delivered a 10-week course for Masters & Undergraduate students in our lab working across Siemens Energy & Pratt & Whitney Canada.
- Designed course content, including weekly lectures, engagement activities, and a project centered on HPT preliminary design and performance optimization

Flight Mechanics & Control (Middle East Technical University)

Dec 2020 - April 2021

- Drop-In review and application sessions facilitated weekly for mechanical and aerospace engineering students, covering flight dynamics, stability and control, aircraft performance, and equations of motion through hands-on exercises in MATLAB and Simulink.

MathWorks (Montreal, Quebec, Canada)

Oct 2021 - Jun 2023

MATLAB Student Ambassador

- Represented MathWorks at École de technologie supérieure (ÉTS).
- Organized and led interactive workshops to showcase MATLAB features and applications.
- Delivered targeted MATLAB training to research groups based on project requirements.
- Partnered with student clubs and university departments to organize events tailored to their engineering disciplines.
- Position created to mirror my previous work with MathWorks during my undergraduate studies at METU.

RELEVANT PROJECTS

Pratt & Whitney Canada Gas Turbine Engine Design

Jan 2022 – May 2022

Concordia University (Montreal, Quebec, Canada)

- Collaborated on preliminary design and development of the HPC and HPT for a 2-shaft geared turbofan engine.
- Developed a MATLAB App Designer optimization tool to support HPT design decisions.
- Conducted trade studies and presented proposed designs at the preliminary design review to a panel of Pratt & Whitney Canada technical experts.

CFD Analysis of VKI Turbine Blade

Sep 2021 – Dec 2021

Concordia University (Montreal, Quebec, Canada)

- Performed CFD analysis of a VKI turbine blade using ANSYS CFX, applying turbomachinery theories to a real-world scenario.
- Generated structured meshes to accurately capture flow dynamics and facilitate post-processing.
- Evaluated simulation performance in modeling shock waves and turbulence effects.

VTOL Personal Aerial Vehicle (PAV) Helicopter Design

Mar 2021 – Jul 2021

Middle East Technical University (Ankara, Turkey)

- Undertook a comprehensive design project focused on creating a PAV helicopter for efficient, customized urban transportation.
- Designed a four-seat, single-engine light utility helicopter for operation in a specific city, considering performance, weight, aerodynamics, and customer appeal.
- Sourced and justified rotorcraft configuration, avionics suite, and control systems for the PAV design.

Long-Range Modern Business Jet Design

Sep 2020 – Feb 2021

Middle East Technical University (Ankara, Turkey)

- Conducted conceptual and preliminary design of a 50-passenger business jet (Paris–NYC, 7.5 hours), focusing on economic and environmental performance.
- Defined mission profile, top-level requirements, and performance targets, and estimated aerodynamics, weight, and stability to finalize sizing and configuration.
- Created CAD models of the aircraft to visualize and refine the design.

Scramjet Engine Design

Sep 2020 – Feb 2021

Middle East Technical University (Ankara, Turkey)

- Inlet and nozzle design of a scramjet operating between Mach 5–15.
- Built a MATLAB application for various operating Mach numbers with 4 ramps. The app shows important performance parameters and the geometry of the inlet (uploaded to MATLAB Add-Ons).

Autopilot Design for Constant Altitude Control

Sep 2020 – Feb 2021

Middle East Technical University (Ankara, Turkey)

- Developed a MATLAB/Simulink model of an aircraft's 6-DOF dynamics to design an autopilot for constant-altitude flight.
- Applied modern control approaches including system linearization, state-space representation, and feedback control.
- Evaluated system stability and performance through simulation-based analysis.

ENGINEERING COMPETITIONS

AIAA Undergraduate Engine Design: Let's Re-Engine The Concorde!

Oct 2020 – Aug 2021

American Institute of Aeronautics and Astronautics (Denver, Colorado, US)

- Designed the turbofan engine "METU-Phoenix" as a candidate replacement for the Rolls-Royce/Snecma Olympus 593 on the Concorde, achieving lower SFC, reduced noise, weight, and size, and increased thrust.
- Performed thermodynamic cycle analysis, performance evaluation, and preliminary sizing to optimize the engine.
- Presented the design at the 2021 AIAA Propulsion And Energy Forum and won 2nd place.
- Link to Report: [Press Here](#)

AIRBUS Sloshing Workshop

Sep 2020 – Jul 2021

AIRBUS & ArianeGroup (Virtual due to COVID)

- Led a 4-member team to design a water rocket demonstrating fuel sloshing reduction, transitioning from rocket launch to glider mode at peak altitude.
- Implemented passive and active control strategies to minimize sloshing effects.
- Conducted CFD analysis and simulations using ANSYS Fluent and MATLAB to evaluate and optimize performance.

AWARDS

Génie+ Student Engagement

Aug 2024

École de technologie supérieure (ÉTS), Montreal, Quebec, Canada

- Mention of honor for my involvement in student life at ETS, recognizing my dedication to scientific, technological, social, and community initiatives through the promotion and realization of student activities.

SARA Exceptional Contribution

Nov 2022

École de technologie supérieure (ÉTS), Montreal, Quebec, Canada

- Award from the "Service d'aide à la Rédaction d'articles" for the innovation and contribution of the Research Grad Chat, a bi-weekly event that helps graduate students collaborate for their research studies.

TECHNICAL SKILLS

- **Programming Languages:** MATLAB (including App Designer & Simulink), Python, C++, SQL, Unix, Perl, Fortran, NPSS, Modelica, Git/GitHub.
- **Engineering Tools & Software:** CAD tools (AutoCad Mechanical, Inventor Professional, SolidWorks, Catia), CAE (ANSYS, AxSTREAM, Modelon Impact).
- **Optimization & Analysis:** Multidisciplinary Design Optimization (MDO / PMDO), surrogate modeling, and data-driven methods (regression, supervised learning, predictive modeling), Machine Learning (ML), Sensitivity & Uncertainty Analysis.
- **Project Management Tools:** Jira, MS Project, Agile, SCRUM

REFERENCES

Available upon request.